Lecture 7: Planetary Surfaces
Geology, Geomorphology, & Geophysics

Planetary Systems

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Olympus Mons, Mars. Credit: NASA/JPL/USGS

Learning objectives

By the end of this lecture, you will be able to:

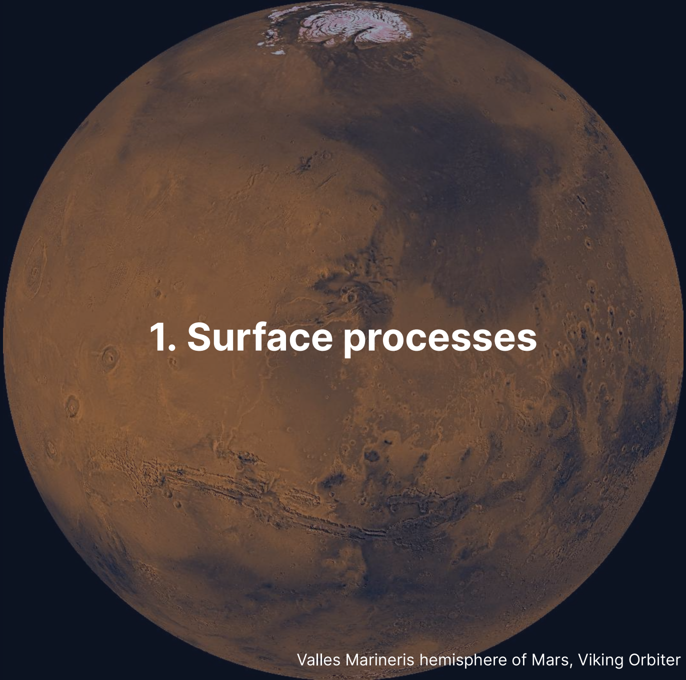
- Distinguish endogenic and exogenic surface processes
- Describe the three stages of impact crater formation
- Derive the crater scaling law $D \sim (E/\rho g)^{1/4}$ from dimensional analysis
- Use crater counting to determine relative and absolute surface ages
- Compare volcanic, tectonic, and erosional styles across the solar system
- Explain cryovolcanism on icy ocean worlds

Recap: Lectures 5 & 6

- Atmospheric structure: hydrostatic equilibrium, scale height, vertical layers
- Energy balance and greenhouse effect: $T_s > T_{\rm eff}$ from IR absorption
- Clausius-Clapeyron equation: cloud formation across the solar system
- Atmospheric dynamics: Hadley cells, Coriolis effect, jet streams
- Carbonate-silicate cycle: Earth's long-term climate thermostat
- Faint young Sun paradox; Venus runaway greenhouse; Mars climate puzzle

Today: What shapes the *surfaces* of planets and moons?

1. Surface processes



Valles Marineris hemisphere of Mars, Viking Orbiter mosaic. Credit: NASA/JPL/USGS

Endogenic vs. exogenic processes

A planet's surface is its geological record: billions of years of competing processes that create, modify, and destroy landforms.

Endogenic (internal heat):

- Volcanism
- Tectonics (faulting, folding)
- More vigorous for larger, hotter bodies

Exogenic (external forces):

- Impact cratering
- Erosion (wind, water, ice)
- Space weathering, radiation

The balance between these processes determines what we see on a surface.

Dominant surface process by body

Body	Dominant process	Key evidence
Moon	Impact cratering	Saturated highland craters, >4 Gyr
Earth	Plate tectonics + erosion	Ocean floors <200 Myr; rapid weathering
Mars	Volcanism + impacts	Olympus Mons; cratered southern highlands
Venus	Volcanism	Volcanic plains cover ~80% of surface
Io	Tidal volcanism	~300–400 active volcanoes; age <1 Myr
Europa	Ice tectonics	Lineae, chaos terrain, very few craters

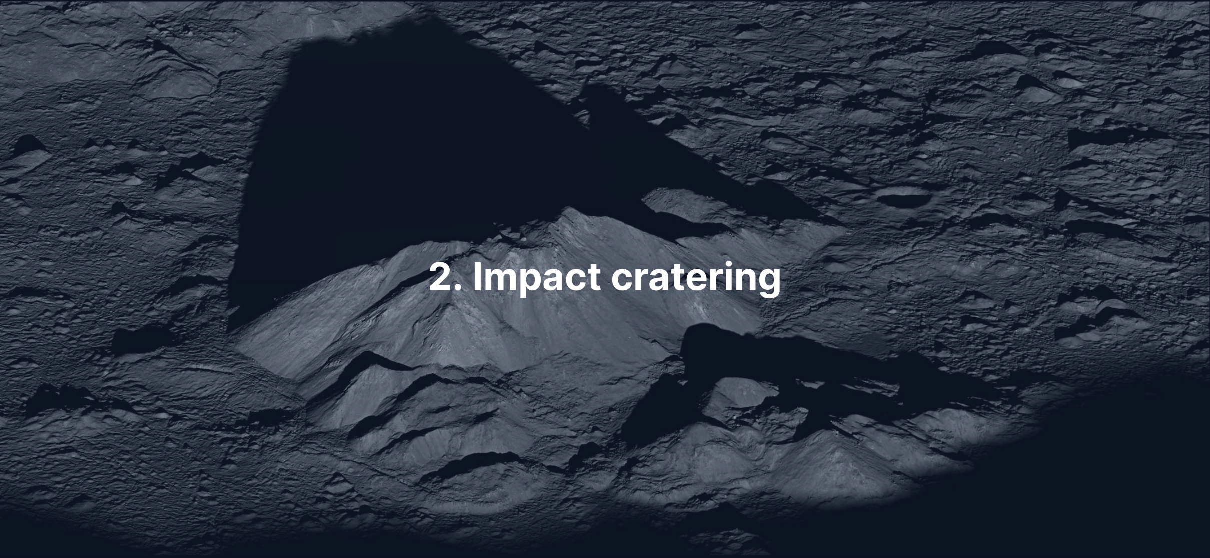
Internal heat → endogenic processes dominate.
Dead interior → surface preserves ancient craters.

Reading the surface record

Credit: NASA/JPL-Caltech/MSSS

- Aeolian dunes sculpted by wind in Gale Crater, Mars
- Wind erosion dominates Mars today
- Ancient volcanism and impacts shaped the deeper record
- On Earth, erosion and plate tectonics erase the crater record instead

2. Impact cratering



Lunar crater morphology. Credit: NASA/GSFC/ASU

The most universal geological process

- **Every** solid body in the solar system bears impact craters
- On bodies without atmospheres or geology (Moon, Mercury), craters survive billions of years
- Impact velocities: 10–70 $\mathrm{km\,s^{-1}}$ (set by orbital mechanics)
- Enormous energy release in a fraction of a second

Three stages of crater formation:

1. Contact and compression
2. Excavation
3. Modification

Stage 1: contact and compression

- Projectile strikes at 10–70 km s^{-1}
- Shock wave propagates into both target and impactor
- Peak pressures ~100 GPa (comparable to Earth's core–mantle boundary)
- Material near impact point is **vaporised and melted**
- Kinetic energy → internal energy in < 1 s

The kinetic energy of an impactor:

$$E_k = \frac{1}{2}mv^2$$

Example: a 1 km asteroid impact

Rocky asteroid: $\rho \approx 3000\ \mathrm{kg\,m^{-3}}$, diameter = 1 km

$$m = \frac{4}{3}\pi r^3 \rho \approx \frac{4}{3}\pi(500)^3 \times 3000 \approx 1.6 \times 10^{12}\ \mathrm{kg}$$

At $v = 20\ \mathrm{km\,s^{-1}}$:

$$E_k = \frac{1}{2} \times 1.6 \times 10^{12} \times (2 \times 10^4)^2 \approx 3 \times 10^{20}\ \mathrm{J}$$

~1000× the energy of the largest nuclear weapon ever detonated (Tsar Bomba, ~50 Mt ≈ 2×10^{17} J), released in <1 s at a single point.

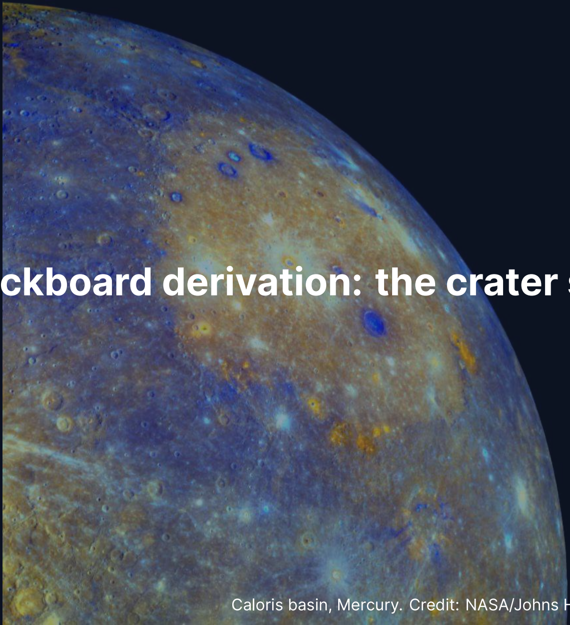
Stages 2 & 3: excavation and modification

Excavation:

- Shock wave expands hemispherically outward
- Rarefaction wave (decompression pulse) accelerates material upward and outward
- **Transient cavity** forms (depth:diameter ≈ 1:3)
- Cone-shaped **ejecta blanket** deposited around the crater

Modification:

- Transient cavity is gravitationally unstable
- Small craters: walls slump inward → **simple crater** (bowl-shaped)
- Large craters: floor rebounds, walls collapse in terraces → **complex crater** with central peak



3. Blackboard derivation: the crater scaling law

Caloris basin, Mercury. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

Setup: crater size from dimensional analysis

Goal: How does crater diameter D depend on impact energy E, target density ρ, and surface gravity g?

In the **gravity regime** (craters ≳100 m, where gravity limits size):

$$D = C\, E^a\, \rho^b\, g^c$$

Dimensions (M, L, T):

$$\begin{aligned} [D] &= L \\ [E] &= ML^2T^{-2} \\ [\rho] &= ML^{-3} \\ [g] &= LT^{-2} \end{aligned}$$

Solving for the exponents

Require $[E^a \rho^b g^c] = L$:

Mass: $a + b = 0 \implies b = -a$

Time: $-2a - 2c = 0 \implies c = -a$

Length: $2a - 3b + c = 1$

Substituting $b = -a$ and $c = -a$:

$$2a + 3a - a = 1 \implies 4a = 1 \implies a = \frac{1}{4}$$

Therefore $a = 1/4$, $b = -1/4$, $c = -1/4$:

$$\boxed{D \sim \left(\frac{E}{\rho g}\right)^{1/4}}$$

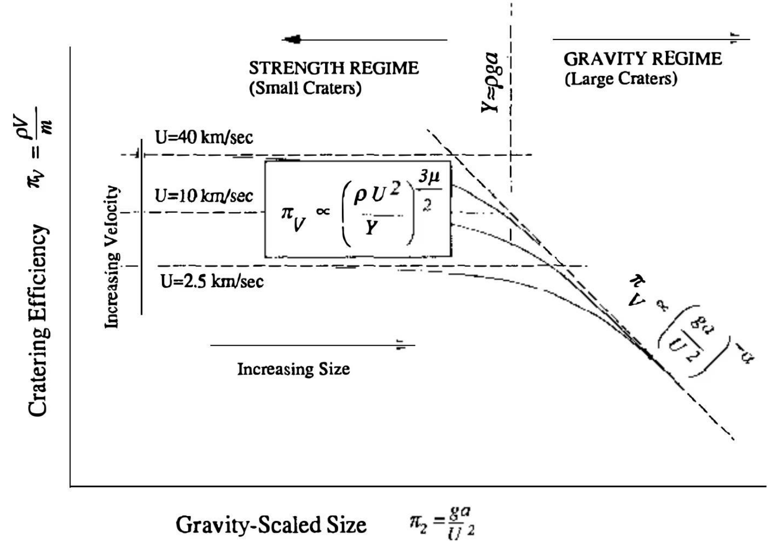
Interpretation of the scaling law

- D scales as the **fourth root** of energy
- Doubling the energy increases D by only $2^{1/4} \approx 1.19$ (19%)
- This explains the relatively narrow range of crater sizes even though impactor energies span many orders of magnitude

Dimensional check:

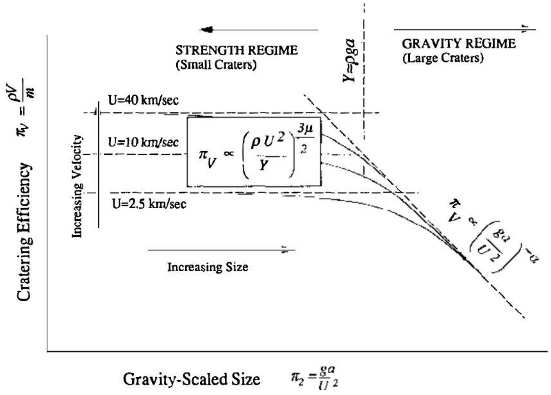
$$\left[\frac{E}{\rho g}\right] = \frac{ML^2T^{-2}}{ML^{-3} \cdot LT^{-2}} = L^4 \quad \Rightarrow \quad [D] = L$$

The more complete **pi-scaling framework** (Holsapple, 1993) parameterises the transition between the gravity and strength regimes.



The two cratering regimes for a target with strength, at impact velocities U = 2.5, 10 and 40 km s^{-1}. Reproduced from Holsapple (1993), Fig. 3.

Where the fourth-root law applies



Credit: Holsapple (1993), Fig. 3

Cratering efficiency $\pi_V = \rho V / m$ against gravity-scaled size $\pi_2 = ga / U^2$:

- **Strength regime** (left): target cohesion limits growth, so π_V depends on impact velocity and the three curves separate
- **Gravity regime** (right): gravity limits growth, the curves converge on one law, $\pi_V \propto \pi_2^{-\alpha}$
- Laboratory impacts sit on the left, planetary craters on the right

Our $D \sim (E / \rho g)^{1/4}$ is the gravity-regime limit. Below ~100 m on the Moon, strength takes over and the law fails.

Worked example: a 1 km asteroid on the Moon

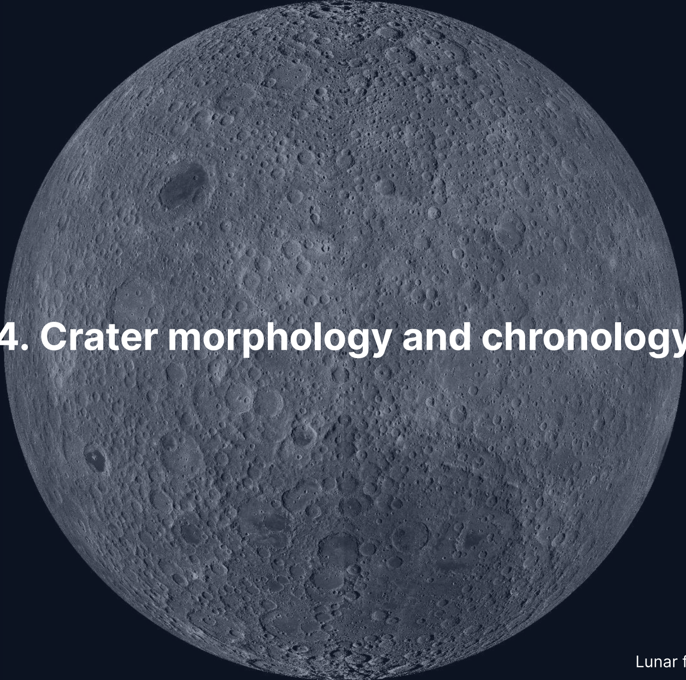
Parameter	Symbol	Value
Impact energy	E	3×10^{20} J
Regolith density	ρ	2500 kg m^{-3}
Surface gravity	g	1.62 m s^{-2}

Regolith: the loose, fragmented debris layer covering the surface.

$$D \sim \left(\frac{3 \times 10^{20}}{2500 \times 1.62}\right)^{1/4} = (7.4 \times 10^{16})^{1/4} \approx 16.5 \text{ km}$$

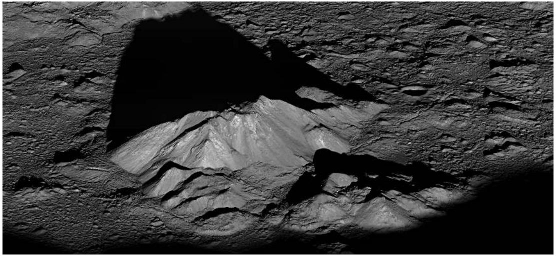
Consistent with observed lunar craters from ~1 km impactors.
Since $D \propto L^{3/4}$ at fixed v and ρ, the 85 km crater Tycho implies a ~9 km impactor (order of magnitude only).

4. Crater morphology and chronology



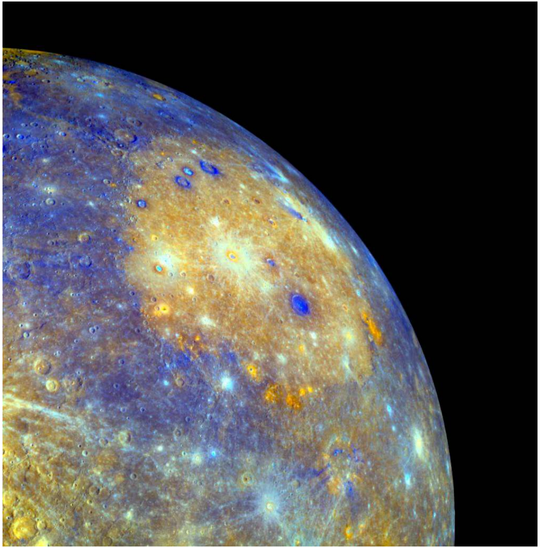
Lunar farside. Credit: NASA/GSFC/ASU

Crater morphology: three classes



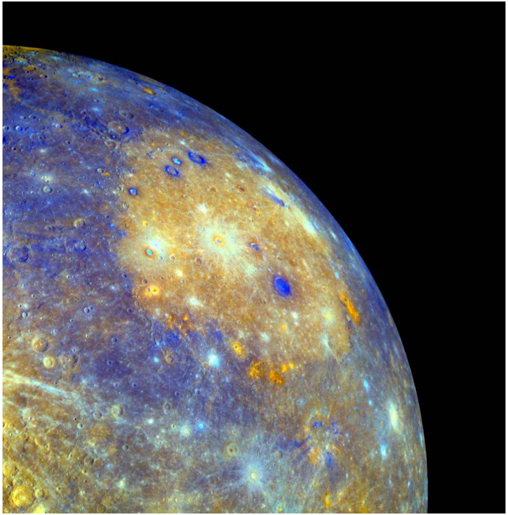
Credit: NASA/GSFC/ASU

1. **Simple** ($D < D_t$)
   Bowl-shaped; depth:diameter ≈ 1:5
   Moon: $D_t \approx 15$ km
2. **Complex** ($D_t < D \lesssim 300$ km)
   Central peak, terraced walls, flat floor
   Gravity overcomes wall strength
3. **Multi-ring basins** ($D \gtrsim 300$ km)
   Concentric ring structures
   Orientale (930 km), Caloris (1550 km), Hellas (2300 km)



The Caloris basin on Mercury in enhanced colour, ~1550 km in diameter. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

Multi-ring basins: Caloris on Mercury



Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

Caloris is ~1550 km across, one of the largest and best preserved multi-ring basins:

- Orange interior plains are smooth volcanic deposits emplaced *after* the impact
- The surrounding annulus holds the ejecta and the concentric rings
- Chaotic “weird terrain” sits at the basin’s antipode, focused there by the seismic shock of the same event

A basin this size is not a scaled-up crater: the ring structure and the antipodal terrain are signatures only the largest impacts produce.

Transition diameter

The simple-to-complex transition scales with surface gravity:

$$D_t \approx D_{t,\mathrm{Moon}} \cdot \frac{g_{\mathrm{Moon}}}{g}$$

Body	g **(m s^{-2})**	D_t **(km)**
Moon	1.62	15
Mars	3.72	~6.5
Earth	9.81	~2.5
Mercury	3.70	~6.6

On Earth, virtually all craters larger than a few km are complex.

Crater size-frequency distribution

The number of craters larger than a given diameter D follows an approximate power law:

$$N(> D) \propto D^{-b}, \quad b \approx 2\text{–}3$$

- Small craters are much more numerous than large ones
- Slope b depends on impactor size distribution and erosion
- A plot of $\log N$ vs. $\log D$ yields a straight line for a pristine surface
- Higher cumulative density $\Rightarrow$ older surface
- Deviations from the power law indicate resurfacing, saturation, or secondary craters

Source: Neukum et al. (2001), *Space Sci. Rev.*

Crater counting: the principle

Key idea: Older surfaces have had more time to accumulate craters.

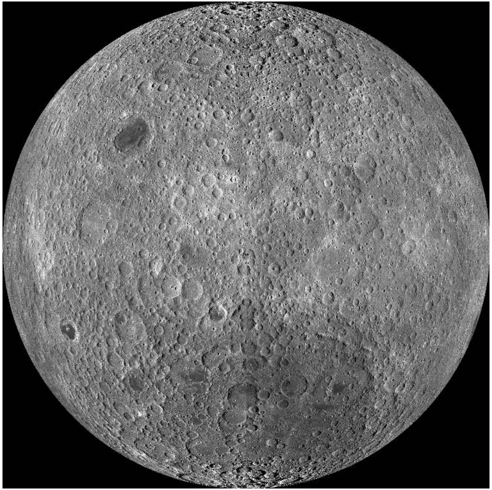
The **crater size-frequency distribution (SFD)** follows a power law:

$$N(> D) = a\,D^{-b}$$

- $N(> D)$: cumulative number of craters per unit area with diameter $> D$
- Production population: $b \approx 2–3$
- Plot N vs. D on log-log axes → straight line
- Higher line = older surface

Crater counting gives **relative ages**. Apollo/Luna samples provide **absolute calibration**.

The lunar chronology



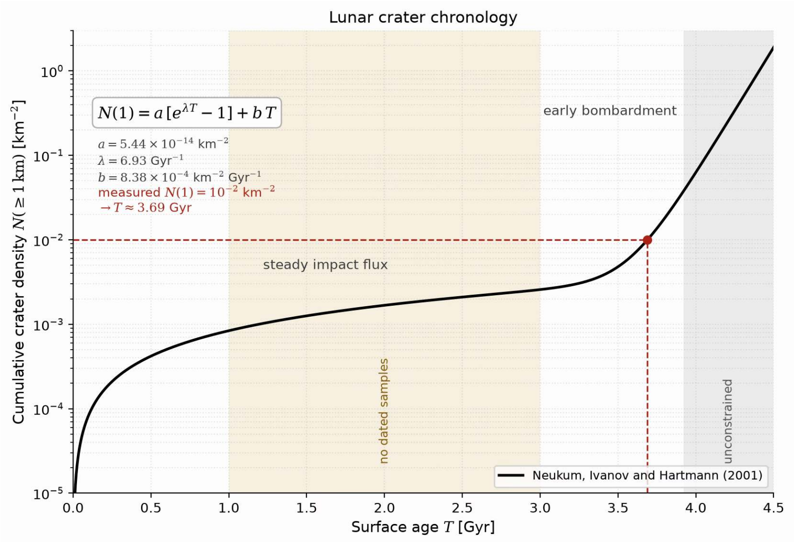
Credit: NASA/GSFC/ASU

Apollo and Luna samples give radiometric ages for specific surfaces:

- **Highlands:** saturated, >4 Gyr
- **Maria:** 3.9–3.1 Gyr
- Calibrated SFD → absolute ages

Extrapolated to other bodies:

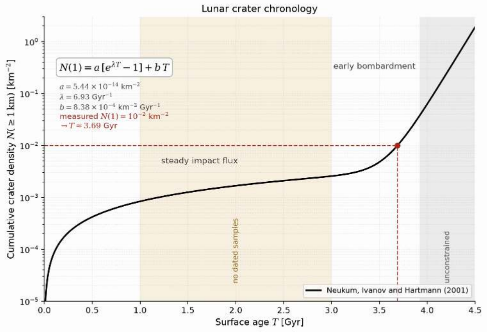
- Mars southern highlands: ~4 Gyr
- Mars northern lowlands: much younger
- Venus: uniform crater density → ~300–700 Myr



The lunar chronology of Neukum, Ivanov and Hartmann (2001), with the intervals that no returned sample constrains. Course figure.

The chronology function

$$N(1) = a\left[e^{\lambda T} - 1\right] + b\,T$$



Credit: Course figure, after Neukum, Ivanov and Hartmann (2001)

- Linear term bT: the steady impact flux of the last ~3 Gyr
- Exponential term: the early bombardment, which overtakes the linear term near 3.6 Gyr
- Read it backwards: a measured $N(1) = 10^{-2}$ km^{-2} gives a surface age of ~3.7 Gyr

The shaded bands hold no dated sample: 1–3 Ga and older than 3.92 Ga. There the curve is interpolation and extrapolation, not calibration.

Key results from crater chronology

- **Late Heavy Bombardment** (~3.9 Ga): spike in impact rate recorded on the Moon
  Caveat: the spike is now discounted. The ~3.9 Ga ages cluster because every Apollo site samples Imbrium ejecta; a smoothly declining flux fits the record (Boehnke & Harrison 2016)
- **Mars hemispheric dichotomy:**
  Southern highlands heavily cratered (~4 Gyr)
  Northern lowlands much smoother → volcanic resurfacing
- **Venus:** Remarkably uniform crater density
  Mean surface age ~300–700 Myr
  Implies a **global resurfacing event**
- **Europa:** Very few craters → surface age ~40–90 Myr
  Continuously resurfaced by ice tectonics

5. Volcanism

Olympus Mons, Mars. Credit: NASA/JPL/USGS

Effusive vs. explosive volcanism

The key variable is **magma viscosity**, controlled by $\mathrm{SiO_2}$ content:

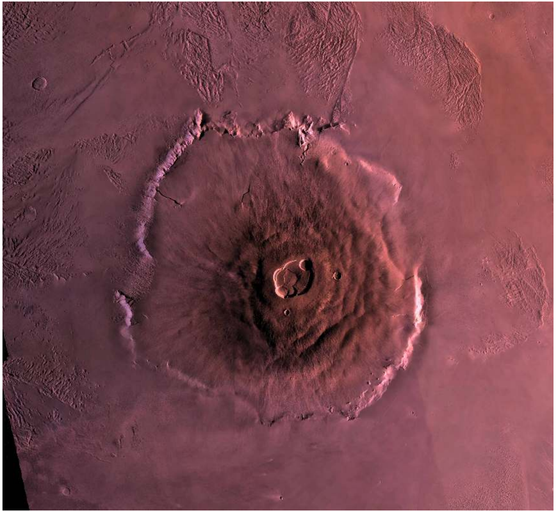
Low viscosity (basaltic):

- ~50% $\mathrm{SiO_2}$
- Flows easily
- Volatiles escape gradually
- **Effusive** eruptions
- Broad shield volcanoes, flood basalts (vast lava plains)
- Moon, Mars, Io

High viscosity (silicic):

- >65% $\mathrm{SiO_2}$
- Traps volatiles
- Pressure builds until failure
- **Explosive** eruptions
- Steep stratovolcanoes (layered lava-ash cones)
- Primarily Earth

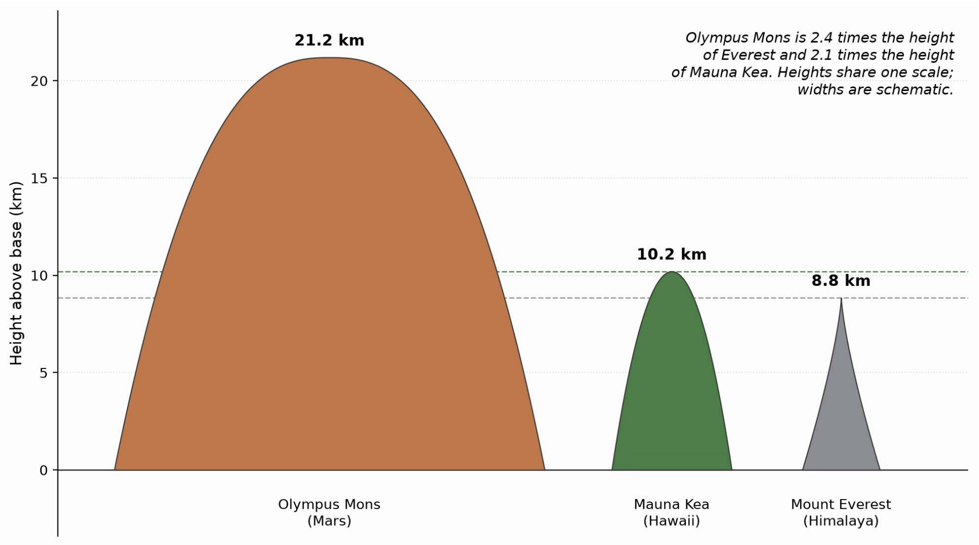
Olympus Mons: the largest volcano in the solar system



Credit: NASA/JPL/USGS

- Shield volcano on Mars
- Base diameter: ~600 km
- Summit: ~21.2 km above the Mars datum
- 2.4× Everest above sea level
- Basal escarpment up to 6 km high

No plate tectonics on Mars → hotspot stays fixed → lava piles up for billions of years in one location. Lower surface gravity ($0.38\,g_{\oplus}$) lets the crust support a taller edifice



Olympus Mons, Mauna Kea and Mount Everest, each rising from its own base. Heights share one linear scale; widths are schematic. Course figure.

How big is Olympus Mons?

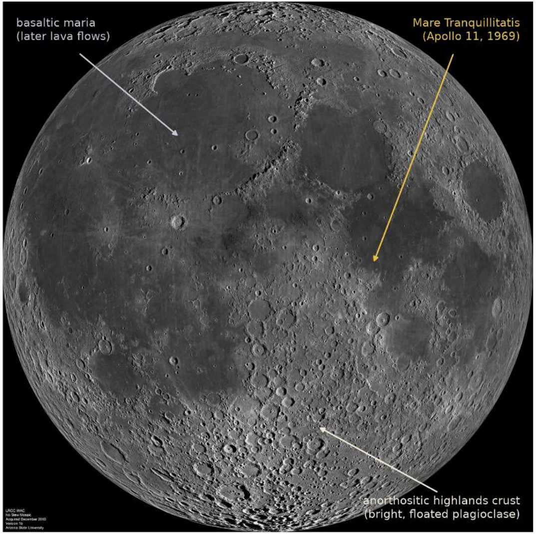
Credit: Course figure

Heights on one linear scale, widths schematic:

- Olympus Mons: ~21.2 km above the Mars datum, ~600 km base
- Mauna Kea: ~10.2 km measured from the ocean floor
- Everest: ~8.8 km above sea level (not a volcano, shown for scale)

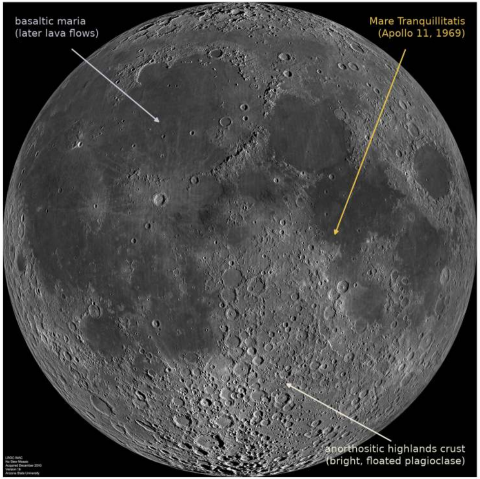
On Earth, plate motion carries a hotspot volcano off its magma source within a few Myr. Surface gravity sets a second limit: maximum edifice height scales as $1/g$, and Mars has $g \approx 0.38\, g_{\oplus}$.

No plate motion: one plume feeds one edifice for billions of years, and the lower gravity allows a ~2.6× taller pile.



The lunar nearside with the major maria labelled. LROC wide-angle mosaic. Credit: NASA/GSFC/Arizona State University.

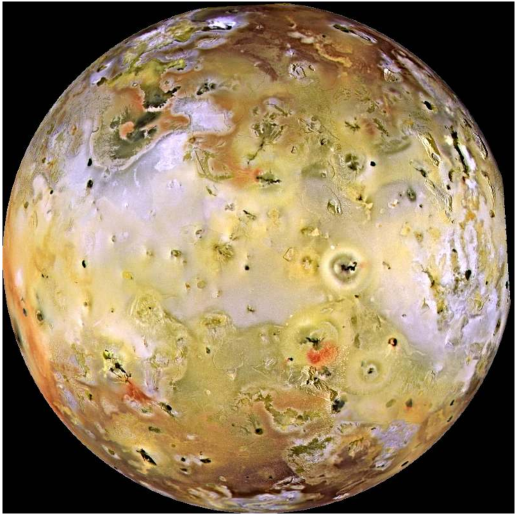
Lunar maria: flood basalts



Credit: NASA/GSFC/Arizona State University

- Vast basalt plains filling ancient impact basins on the nearside
- Erupted between 3.9 and 3.1 Ga (radiometric ages from Apollo samples)
- Cover ~16% of the lunar surface
- Visible from Earth as the dark patches forming the “face” of the Moon
- Record a period of residual internal heating, now extinct

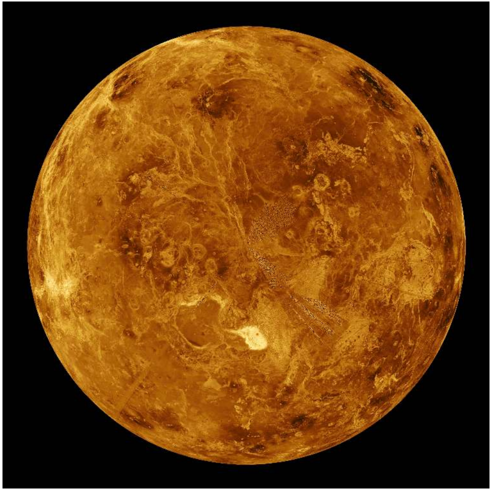
Io: the most volcanically active world



Credit: NASA/JPL-Caltech/University of Arizona

- Most volcanically active body in the solar system
- ~300–400 active volcanic centres
- Powered by tidal heating (Laplace resonance with Europa and Ganymede)
- Surface age <1 Myr: no impact craters survive

Venus under the clouds



Credit: NASA/JPL

Magellan (1990–1994) mapped 98% of the surface at ~100 m resolution, through a permanent cloud deck:

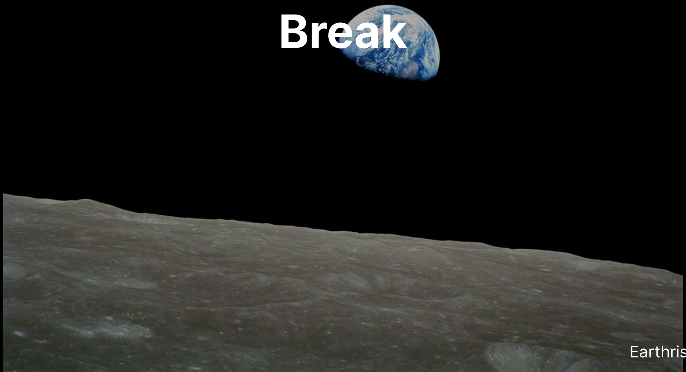
- More than 1600 volcanic centres
- Lava plains over ~80% of the surface
- A stagnant lid (immobile surface shell); possible episodic overturns
- A uniform crater population: one mean surface age of 300–700 Myr

Everything we know of Venus geology rests on one radar mission, which is why VERITAS and EnVision matter.

Volcanism across the solar system

Body	Style	Driving mechanism	Examples
Earth	Effusive + explosive	Internal heat + recycling	Hawaii, mid-ocean ridges
Moon	Flood basalts (extinct)	Residual heat (3.9–3.1 Ga)	Maria
Mars	Shield volcanoes	Mantle plumes, no plates	Olympus Mons, Tharsis
Venus	Lava plains + shields	Internal heat, stagnant lid	Maat Mons
Io	Ultramafic flows	Tidal heating	Loki Patera, Pele

Break

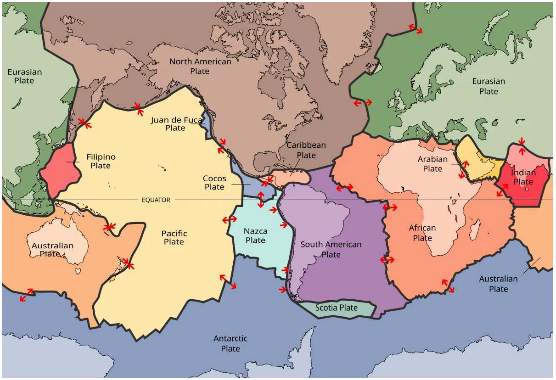


Earthrise, Apollo 8 (1968). Credit: NASA

6. Tectonics

The San Andreas Fault across the Carrizo Plain, California. Credit: Ikluft, Wikimedia Commons, CC BY-SA 4.0

Plate tectonics: Earth's unique regime



Credit: Wikimedia, CC BY-SA 3.0

Earth is the **only** body with active plate tectonics:

- ~15 rigid plates, moving at 1–10 cm yr^{-1}
- **Divergent:** mid-ocean ridges (new crust)
- **Convergent:** subduction zones (recycling)
- **Transform:** horizontal sliding

Enables the **carbonate-silicate cycle**:
Carbon recycling → long-term climate regulation.

Stagnant-lid convection

- All other terrestrial bodies: **single rigid lithospheric lid**
- Mantle convects beneath, but lid does not participate
- Heat escapes by conduction through the lid and occasional volcanism
- Lid thickens over time as interior cools → volcanism eventually shuts down

Why is Earth unique?

- Viscosity contrast at lid base: ~ 10^{10}
- Breaking the lid requires water weakening + self-sustaining damage
- Likely a combination of size, water content, and thermal history

Mars: Tharsis and Valles Marineris

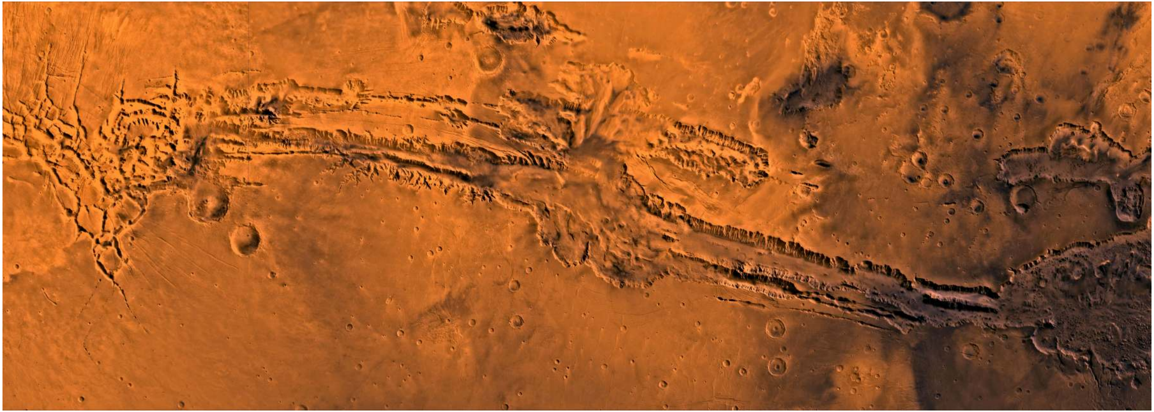
Credit: NASA/JPL/USGS

Tharsis bulge:

- Volcanic plateau ~5000 km across, ~10 km high
- Likely uplifted by a mantle plume (rising column of hot mantle rock)

Valles Marineris:

- Largest canyon system: ~4000 km long
- Up to 7 km deep, 200 km wide
- Extensional rifting from Tharsis
- Lisbon to Moscow in scale



Valles Marineris in a Viking Orbiter mosaic: Noctis Labyrinthus on the left, then the Melas, Candor and Coprates chasmata, and the chaotic terrain feeding Chryse Planitia on the right. Credit: NASA/JPL/USGS.

Valles Marineris

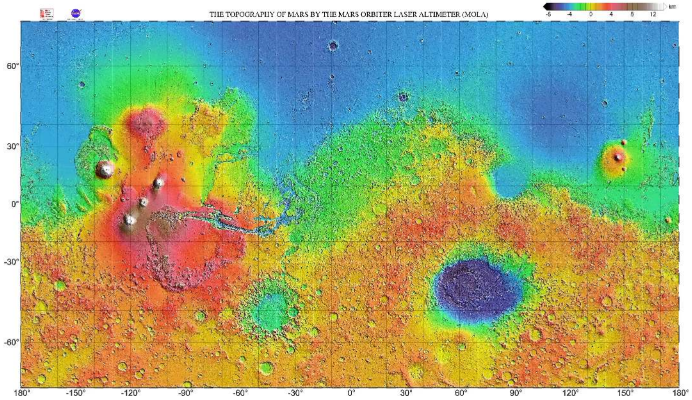
Credit: NASA/JPL/USGS

The largest canyon system in the solar system:

- ~4000 km long, up to 7 km deep, 200 km wide
- Formed by extensional rifting tied to the Tharsis bulge to the west
- Widened afterwards by mass wasting and possibly by water

Mars has no plate boundaries, yet it has the solar system's largest rift. Loading by one volcanic province did what plate divergence does on Earth.

Mars topography: the hemispheric dichotomy



Credit: NASA/JPL/GSFC/MOLA Science Team

~6 km elevation difference between southern highlands and northern lowlands. Tharsis bulge with 4 giant volcanoes; Hellas basin (−8.2 km); Valles Marineris at equator.

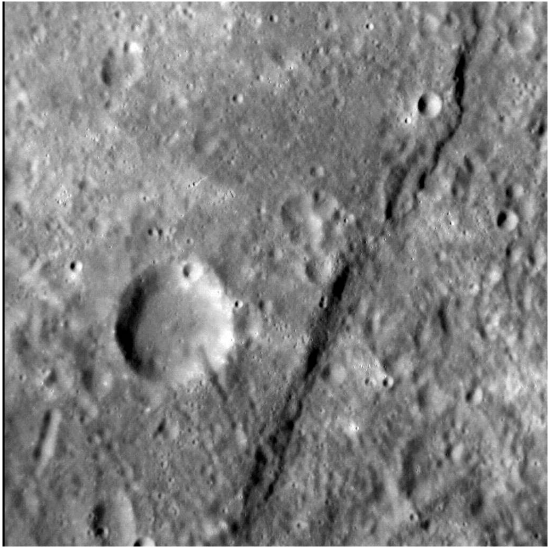
Venus and Mercury: contrasting tectonic styles

Venus, episodic resurfacing:

- Uniform crater density → mean age ~300–700 Myr
- Hypothesis: periodic catastrophic lid foundering
- Entire surface resurfaced volcanically in a short interval
- Between episodes: tectonically quiet

Mercury, global contraction:

- Large iron core cooled and solidified over time
- Planet's radius shrank by ~7 km
- Produced **lobate scarps**: thrust faults up to several hundred km long, 1–3 km high
- Mapped by Mariner 10 and MESSENGER



A lobate scarp on Mercury, a contraction thrust fault, imaged by MESSENGER. Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington.

Mercury: a planet that shrank

Credit: NASA/Johns Hopkins APL/Carnegie Institution of Washington

A lobate scarp near Pourquoi-Pas crater, the surface expression of a thrust fault:

- Mercury's large iron core cooled and solidified, shrinking the radius by up to ~7 km
- The crust took up that contraction along thrust faults hundreds of kilometres long and 1–3 km high
- Mariner 10 found them; MESSENGER mapped them globally

The fault breaks terrain that is already cratered, so the contraction is younger than the craters it cuts. A cooling interior keeps deforming a stagnant lid.

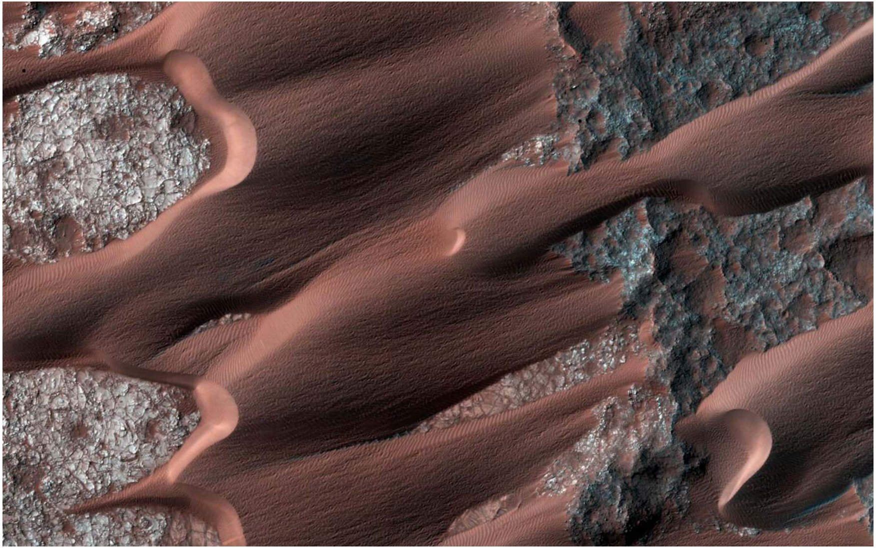
7. Erosion and weathering

Barchan dunes in Nili Patera, Mars, imaged by HiRISE. Credit: NASA/JPL-Caltech/University of Arizona

Aeolian (wind) processes

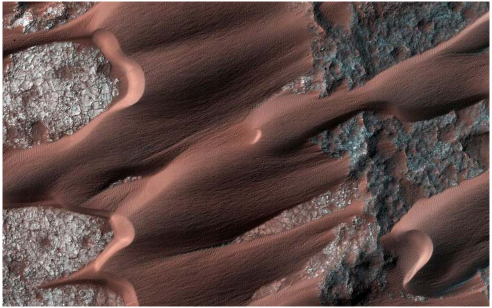
Wind erosion and deposition require a sufficiently dense atmosphere:

- **Mars:** Extensive dune fields in craters and polar regions
  Global dust storms loft particles to 30 km altitude
  Active dune migration observed by Curiosity
- **Titan:** Vast equatorial dune fields of organic particles (tholins)
  Longitudinal dunes up to 150 m tall, hundreds of km long
- **Venus:** Dense atmosphere but slow surface winds (~1 $\mathrm{m\,s^{-1}}$)
  Even slow winds can mobilise fine particles in thick atmosphere



Barchan dunes in Nili Patera, Mars, from HiRISE on the Mars Reconnaissance Orbiter. Credit: NASA/JPL-Caltech/University of Arizona.

Sand that still moves



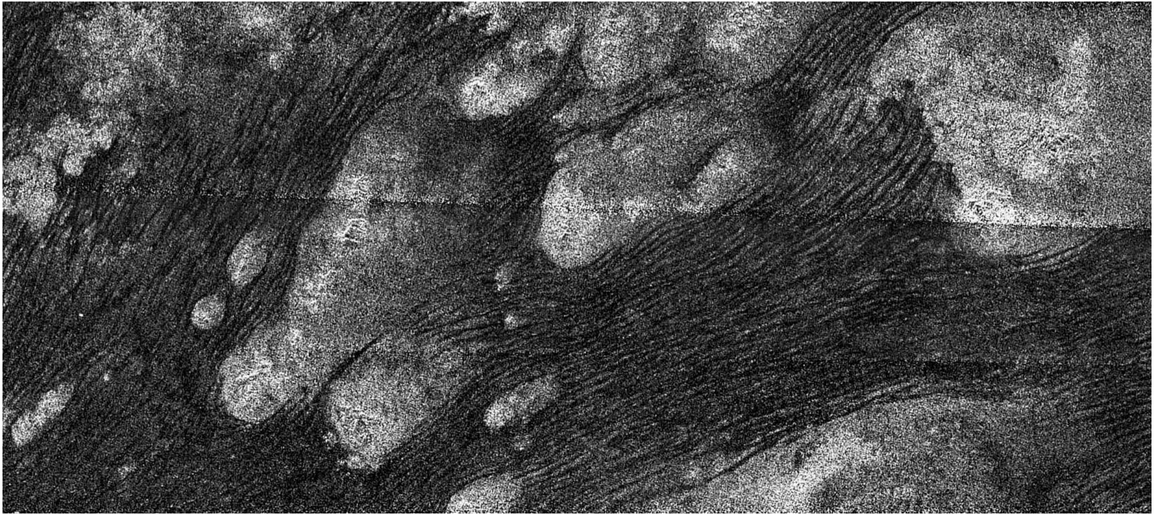
Credit: NASA/JPL-Caltech/University of Arizona

A barchan dune field inside the Nili Patera caldera (a volcanic collapse crater), imaged repeatedly by HiRISE:

- Mars's surface pressure is ~600 Pa, about 170 times thinner than Earth's
- Time-resolved imaging still measures sand fluxes comparable to terrestrial ones
- Repeat images show whole dunes advancing, not only the ripples on them

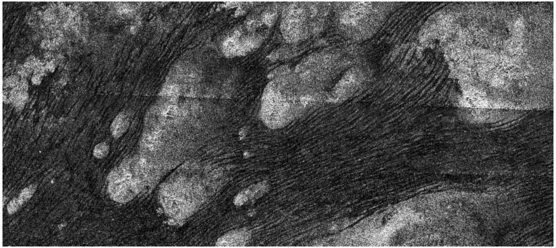
A thin atmosphere does not mean a dead surface. Once grains are moving, low density is partly offset by the low gravity and the long saltation hops.

Source: Bridges et al. (2012), *Nature*



Longitudinal dunes of organic sand in Titan's Shangri-La region, Cassini SAR. Credit: NASA/JPL-Caltech/ASI.

Titan’s organic dunes



Credit: NASA/JPL-Caltech/ASI

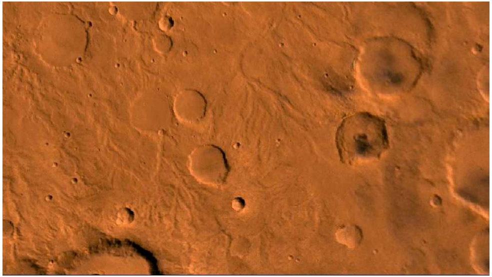
Cassini radar of the Shangri-La dune fields near Titan’s equator:

- Dark parallel ridges are dunes of tholins, organic grains that settle out of the atmosphere
- Up to ~150 m tall and hundreds of kilometres long
- Bright patches are water-ice bedrock that the sand flows around

Same landform, different chemistry: the grains are organic and the bedrock is ice, yet the dune shapes follow the same aeolian physics as on Earth and Mars.

The Colorado River in the Grand Canyon: ~1.6 km of incision in 5–6 Myr, the type example of fluvial erosion on Earth. Credit: Alex Demas/USGS.

Fluvial (water) processes



Credit: NASA/JPL/USGS

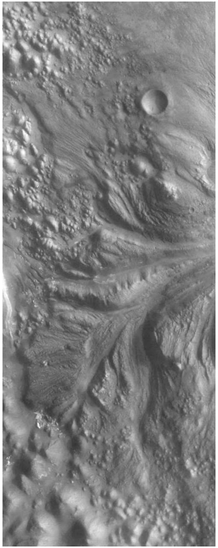
Mars valley networks:

- Noachian-aged (Mars's oldest geologic period, >3.7 Ga)
- Dendritic drainage patterns resembling terrestrial rivers
- Evidence for sustained liquid water flow

Outflow channels:

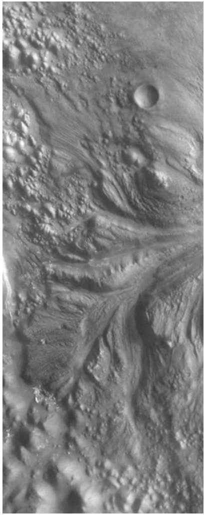
- Hundreds of km long
- Catastrophic groundwater releases

Titan: methane rivers; Huygens found rounded ice pebbles in a dry riverbed (2005).



A Martian outflow channel cutting through Aram Chaos. Credit: NASA/JPL-Caltech/MSSS.

Two kinds of water feature on Mars



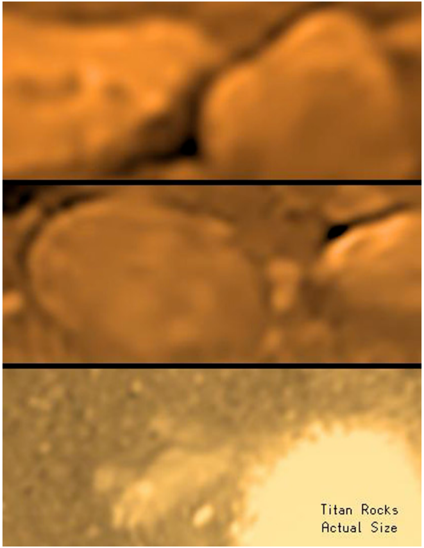
Credit: NASA/JPL-Caltech/MSSS

Outflow channels (shown here, Aram Chaos):

- Braided streamlined islands, broad scoured troughs
- Tens of kilometres wide, hundreds long
- Catastrophic discharge, plausibly a sudden release of groundwater or subsurface ice

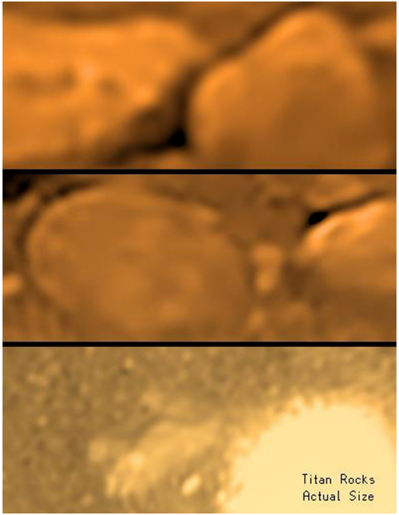
Valley networks are the other kind: dendritic, low discharge, sustained for long enough to need precipitation.

Read the shape, not just the presence of water. One says flood, the other says climate.



The Huygens surface image, in three progressive enlargements; the bottom panel is at the true apparent size. Credit: ESA/NASA/JPL-Caltech/University of Arizona.

Standing on Titan



Credit: ESA/NASA/JPL-Caltech/University of Arizona

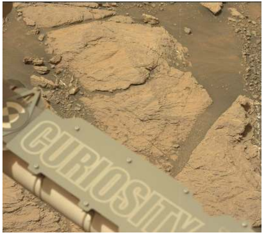
The Huygens probe landed on 14 January 2005, the only surface image ever returned from the outer solar system:

- Rounded decimetre-scale blocks of water ice, shaped by methane flow
- A damp, dark plain of methane-soaked organic sediment in a dried-out riverbed
- Descent images showed dendritic drainage above it

Titan runs the only active fluvial cycle beyond Earth. The liquid is methane, the bedrock is ice, and the rounding of the cobbles is the same process as in a terrestrial stream.

Glacial and chemical weathering

Fedchenko Glacier, Pamir. Credit: NASA/ISS



Clay-bearing unit, Glen Torridon, Gale crater. Credit:

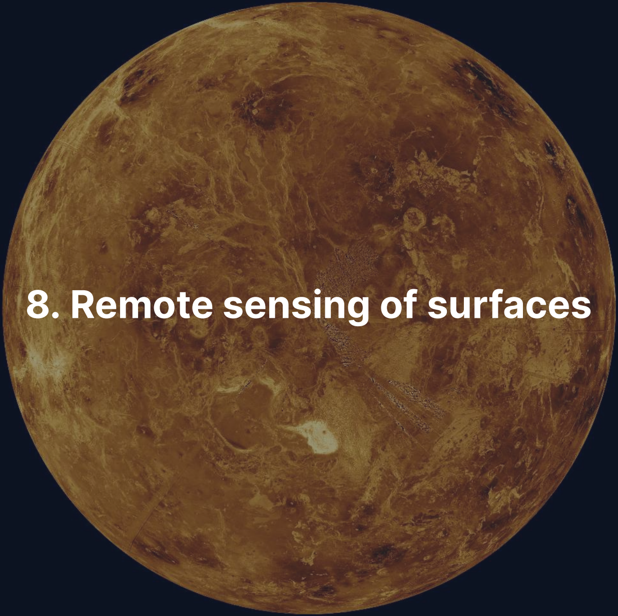
Glacial processes:

- Earth: ice sheets covered ~30% of land at Last Glacial Maximum (~20 ka)
- Mars: polar $\mathrm{CO_2}$ and $\mathrm{H_2O}$ ice caps; mid-latitude features resemble rock glaciers

Chemical weathering:

- Earth: silicate weathering by carbonic acid is the critical carbon sink
- Mars: orbital spectroscopy (CRISM, OMEGA) detected hydrated minerals: phyllosilicates, sulfates, carbonates
- Venus: high surface T (~737 K) drives surface–atmosphere reactions

8. Remote sensing of surfaces



Venus in Magellan radar, the surface seen through an opaque atmosphere. Credit: NASA/JPL

Four ways to see a surface from orbit

We can visit only a few sites. Most surface knowledge comes from orbital remote sensing.

Reflectance spectroscopy (UV–IR) identifies minerals via absorption bands

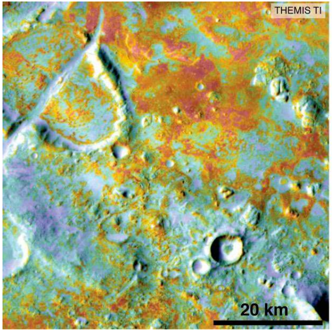
Radar imaging penetrates clouds and regolith, measures surface roughness

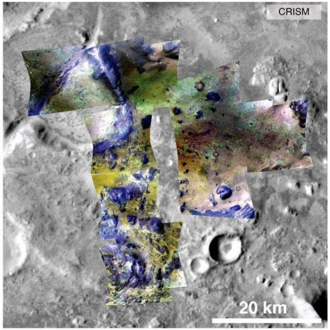
Laser altimetry maps topography at sub-metre vertical precision

Gravity field mapping constrains subsurface density anomalies

Each technique probes a different length scale and physical property; together they build a three-dimensional picture of the surface and shallow interior.

Reflectance spectroscopy: mineral mapping





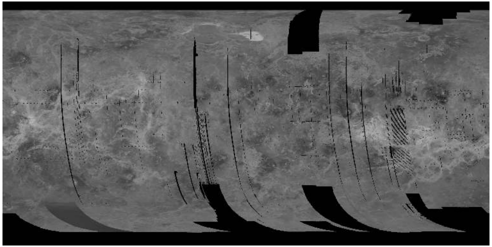
Credit: NASA/JPL-Caltech/ASU/JHU APL

Nili Fossae, Mars:

- Left: THEMIS thermal inertia
- Right: CRISM near-IR mineralogy
- Green: Mg-carbonates
- Yellow: olivine; blue: basalt

Strongest mineralogical evidence for past liquid water on Mars.

Radar imaging: seeing through clouds



Credit: NASA/JPL Magellan Cycle 1

Active microwave imaging is insensitive to clouds and darkness.

Magellan at Venus (1990–94):

- SAR mapped 98% of the surface at ~100 m resolution
- Coronae, tesserae (deformed terrain), lava flows, young surface

Cassini at Titan (2004–17):

- Ku-band radar pierced the organic haze
- Mapped hydrocarbon lakes and seas

Radar is the only way to image surfaces shrouded by thick atmospheres.

Laser altimetry and gravity mapping

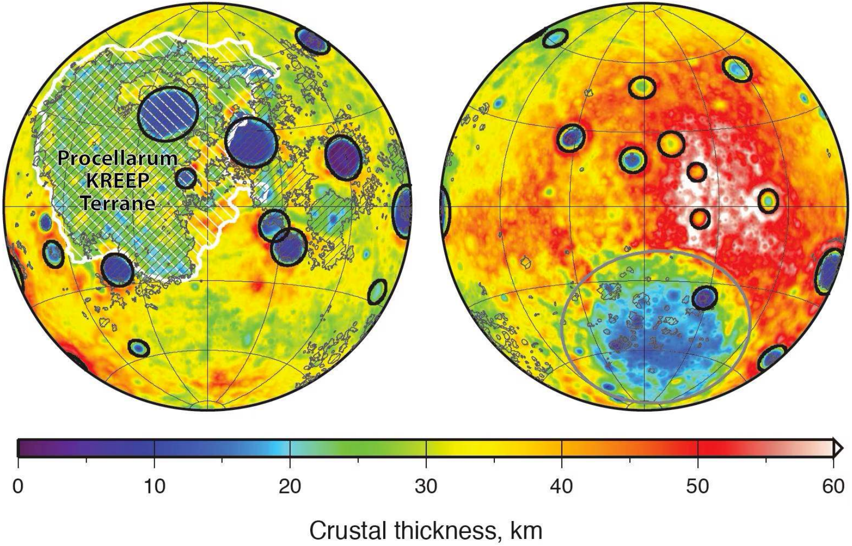
Laser altimetry (MOLA, LOLA, BELA):

- Time-of-flight of laser pulses from orbit
- Vertical precision ~1 m over global baselines
- MOLA produced the canonical Mars topographic map (used throughout this lecture)

Gravity field mapping (GRAIL at the Moon, GRACE/GOCE at Earth):

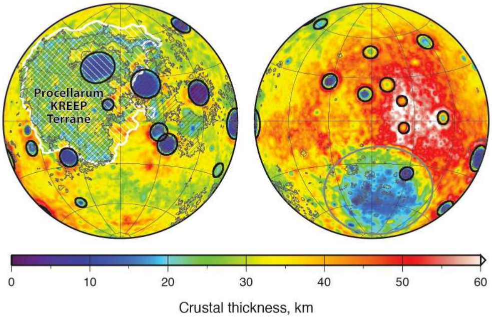
- Spacecraft-to-spacecraft tracking measures tiny accelerations
- Gravity anomalies reveal buried mass: dense cores, thin crust, basins, mantle plumes
- Combined with topography, constrains crustal thickness and compensation state

Altimetry and gravity together tell us what the interior **looks like** from above.



Lunar crustal thickness from GRAIL, nearside (left) and farside (right). Credit: NASA/MIT/GSFC/JPL-Caltech.

Weighing the lunar crust



Credit: NASA/MIT/GSFC/JPL-Caltech

GRAIL tracked the distance between two co-orbiting satellites to map the Moon's gravity field:

- Gravity plus topography gives the depth to the crust-mantle interface
- Mean crustal thickness ~34–43 km
- Strong thinning under the largest basins (deep blue), a thicker farside crust

Gravity sees what images cannot: an impact basin is not just a surface scar, it is a hole punched through most of the crust.

9. Regolith and space weathering

Apollo 11 bootprint in the lunar regolith. Credit: NASA

A shattered skin on every airless world

Credit: NASA/Apollo 11

Regolith:

- Loose, fragmented surface layer on airless bodies
- Produced by impact gardening (repeated impact mixing)
- Depth: 2–4 m maria, 6–8 m highlands

Space weathering:

- Solar wind ion implantation
- Micrometeorite melting → nanophase iron (tiny metallic iron grains)
- Surfaces become **darker and redder**
- Fresh craters appear bright (e.g., Tycho)

10. Cryovolcanism on icy bodies

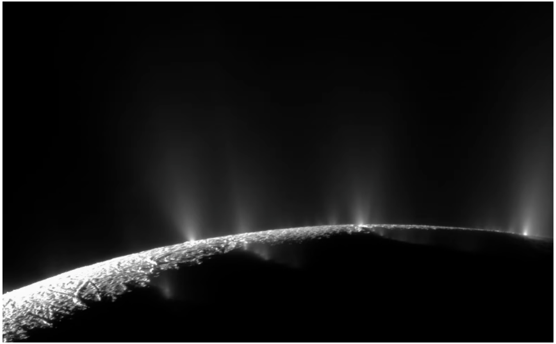
Enceladus plumes. Credit: NASA/JPL-Caltech/SSI

Cryovolcanism: volcanism on ice worlds

In the outer solar system, “volcanic” eruptions involve volatiles rather than silicate melts:

- Erupted materials: liquid water, ammonia–water mixtures, methane, nitrogen
- Energy source: **tidal heating** maintains subsurface oceans beneath icy shells
- “Magma” = low-viscosity volatile liquids
- Analogous to silicate volcanism, but at temperatures of 100–270 K

Enceladus: tiger stripes and geysers



Credit: NASA/JPL-Caltech/SSI (Cassini)

- Radius only 252 km
- Water ice jets from 4 parallel "tiger stripe" fractures
- Sourced from **global subsurface ocean**
- Ocean in contact with rocky core
- Thermal emission: ~15.8 GW

Plume composition:

- H_2, silica grains, organics
- Consistent with hydrothermal vents

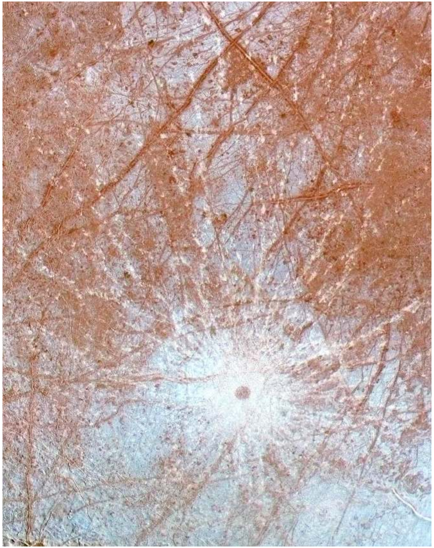
Europa: a hidden ocean

- Jupiter's moon, $R \approx 1561$ km
- Global ocean ~100 km deep; ice shell ~15–25 km thick
- Maintained by tidal heating (Laplace resonance with Io and Ganymede)

Surface features:

- **Lineae:** cracks filled by upwelling ocean water or warm ice
- **Chaos terrain:** broken, rotated, refrozen ice blocks
- **Very few craters:** surface age ~40–90 Myr

Europa Clipper (launched 2024): dozens of close flybys to characterise the ice shell, ocean, and habitability.



Double ridges and lineae near Pwyll crater on Europa, Galileo SSI. Credit: NASA/JPL-Caltech/University of Arizona/University of Colorado.

Europa’s cracked shell

Credit: NASA/JPL-Caltech/University of Arizona/University of Colorado

Galileo imaged the region around Pwyll crater, the bright ray system at lower right:

- Dark double ridges and lineae criss-cross the whole surface
- Each set records one episode of fracturing as tidal stress flexed the brittle ice
- Chaos terrain marks where the shell broke up, rotated and refroze

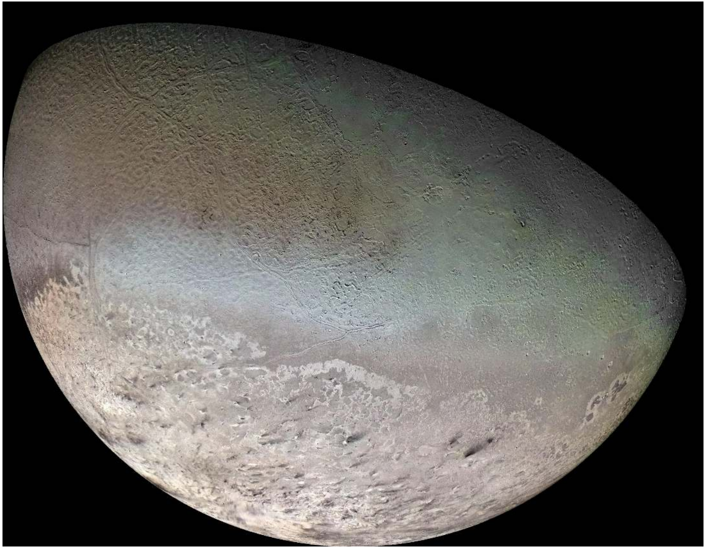
A crater density this low means a surface age of only 40–90 Myr. On a 4.5 Gyr-old moon, that is resurfacing happening now.

Triton and the ocean worlds

Triton (Neptune's largest moon):

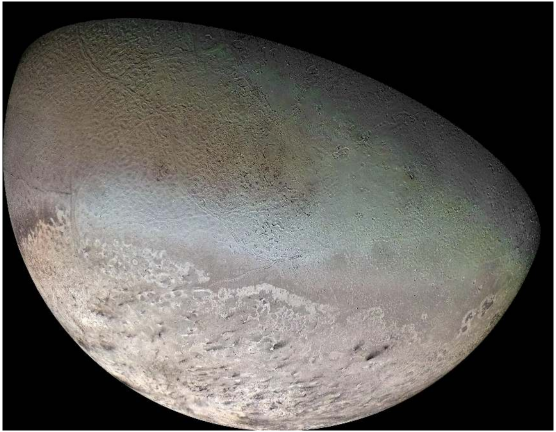
- Nitrogen geysers observed by Voyager 2 (1989)
- Plumes of N_2 gas and dark dust, rising ~8 km
- Very young surface; retrograde orbit (captured KBO)
- Possible subsurface ocean

Enceladus, Europa, and Triton demonstrate that liquid water can exist far beyond the classical habitable zone, maintained by tidal heating rather than stellar radiation.



Voyager 2 colour mosaic of Triton, 1989. Credit: NASA/JPL/Voyager 2.

Triton's nitrogen geysers



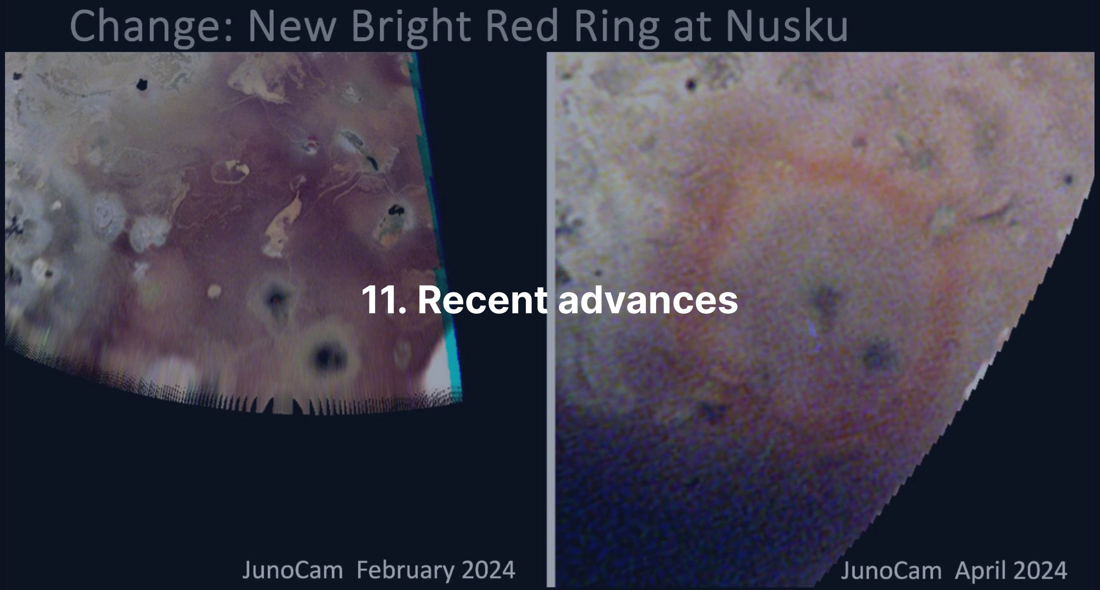
Credit: NASA/JPL/Voyager 2

Voyager 2 flew past in 1989 and found an active surface:

- A pinkish southern polar cap of N_2 and CH_4 frost
- Dark streaks are dust carried downwind from nitrogen geysers rising ~8 km
- North of the cap, "cantaloupe terrain" that has no analogue anywhere else

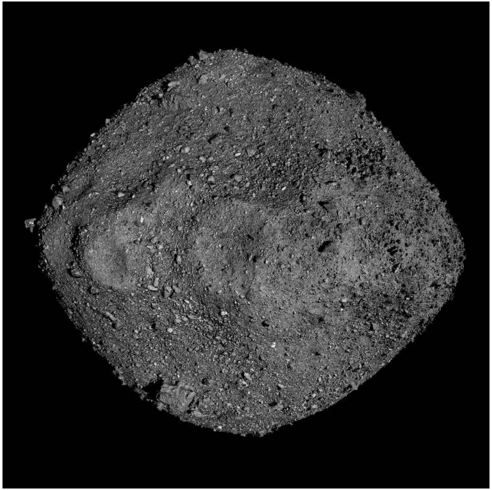
Triton orbits retrograde, so it is a captured Kuiper-belt object. Its active surface is our closest look at what a Pluto-class body does when it is warmed.

11. Recent advances



A new eruption on Io, JunoCam 2024. Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry

OSIRIS-REx: rubble-pile asteroids



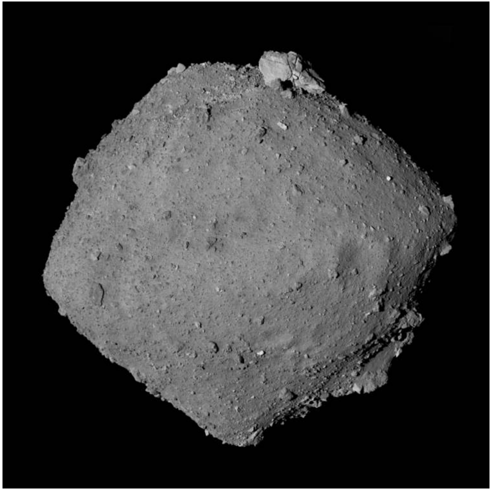
Credit: NASA/GSFC/University of Arizona

NASA's OSIRIS-REx returned 121.6 g of asteroid **Bennu** in September 2023.

- ~500 m rubble-pile asteroid
- Near-Earth orbit, carbonaceous (B-type), water-rich
- Hydrated clays, carbonates, phosphates
- Amino acids and nucleobases
- Extremely porous: a "loose pile of rocks"
- Insight into early solar system aqueous alteration

Source: Lauretta et al. (2019, 2024)

Hayabusa2: sampling Ryugu

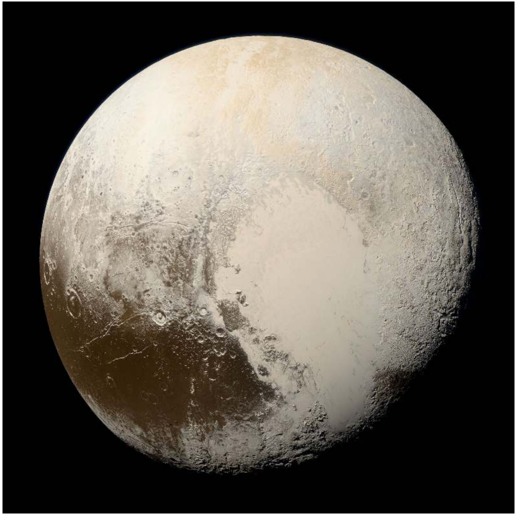


Credit: JAXA, University of Tokyo and collaborators

JAXA's Hayabusa2 delivered 5.4 g of asteroid Ryugu to Earth in December 2020.

- C-type carbonaceous asteroid, 900 m diameter
- First sample of a water-bearing asteroid returned to Earth
- Contains pristine organic molecules: amino acids, sugars, uracil
- Evidence for aqueous alteration on the parent body
- Noble gas and rare-earth abundances constrain its formation location

Pluto: a surprisingly active world



Credit: NASA/JHUAPL/SwRI (New Horizons 2015)

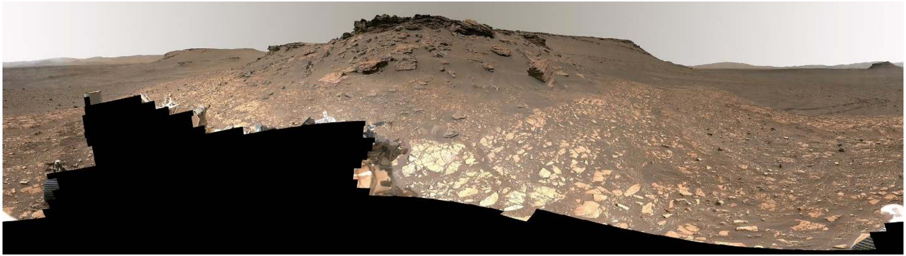
New Horizons (2015) revealed Pluto's geological diversity.

- **Sputnik Planitia:** 1000 km nitrogen ice glacier
- Cellular convective patterns: active N_2 ice convection today
- Surface age ≲ 10 Myr
- Mountains of water ice up to 3 km tall
- Possible subsurface liquid water ocean

Source: Stern et al. (2015); Moore et al. (2016)

Perseverance at Jezero crater

- Operating since February 2021 in Jezero, an ancient lake crater with a preserved delta deposit
- Crater floor: olivine-bearing igneous rock, later altered by water
- >20 sample tubes cached for Mars Sample Return
- Joint NASA/ESA mission: first laboratory analysis of Martian rocks, targeting past habitability and possible biosignatures



The Jezero delta front, Mastcam-Z enhanced colour. Credit: NASA/JPL-Caltech/ASU/MSSS

DART: a planetary defence experiment

Credit: NASA/Johns Hopkins APL

- **Double Asteroid Redirection Test** (September 2022)
- Kinetic impactor deliberately crashed into moonlet **Dimorphos**
- Changed orbital period around Didymos by **33 minutes**
- Confirmed kinetic impact as a viable deflection strategy
- Impact ejecta revealed properties of rubble-pile asteroid surfaces

We can deflect a hazardous asteroid with existing technology.

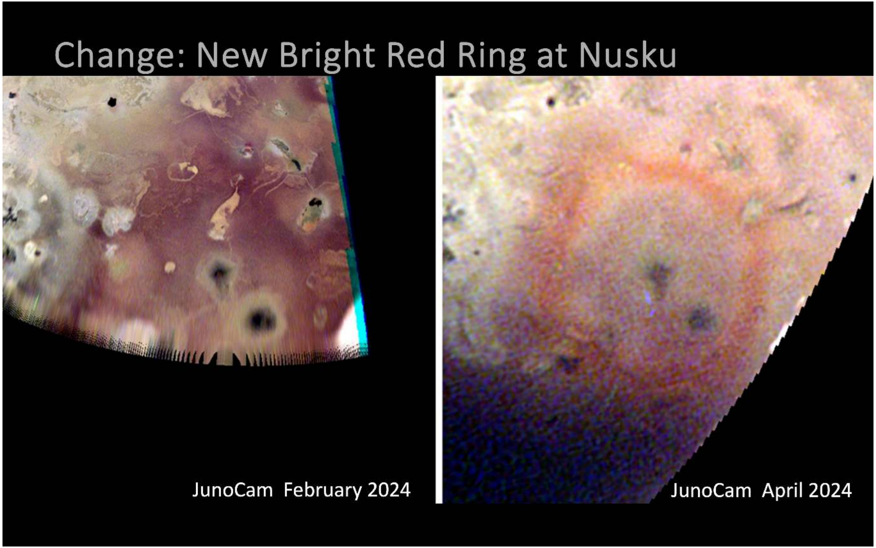
Ongoing exploration: Io and Venus

Io monitoring:

- Juno extended mission: close Io flybys in 2023–2024
- Discovery of previously unknown eruption sites
- Constraints on spatial distribution of tidal heat flow
- Ground-based adaptive optics complement spacecraft data

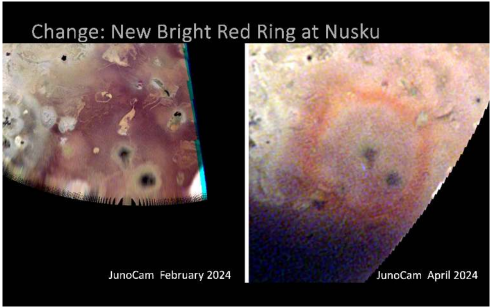
Upcoming Venus missions:

- VERITAS and EnVision (planned for early 2030s)
- First high-resolution surface radar data since Magellan (1990–1994)
- Will test whether Venus has recent or ongoing volcanic activity



The Nusku region on Io two months apart in 2024, JunoCam. Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry.

Catching an eruption in the act on Io



Credit: NASA/JPL-Caltech/SwRI/MSSS/Jason Perry

JunoCam imaged the Nusku region on Io twice, two months apart in early 2024:

- February: nothing unusual around the vent
- April: a bright red ring of fresh sulfur-rich pyroclastic deposits
- One eruption, recorded between two flybys

Io is the only body where we can watch surface geology change on a timescale of weeks. Juno gives the first sustained monitoring since Galileo.

Summary

1. Surfaces record the competition between **endogenic** (volcanism, tectonics) and **exogenic** (impacts, erosion) processes
2. Impact cratering is universal; three stages: contact, excavation, modification
3. Crater scaling law: $D \sim (E/\rho g)^{1/4}$ (fourth-root dependence on energy)
4. Crater counting + Apollo calibration → absolute surface ages
5. Volcanism style depends on magma viscosity; no plate tectonics → giant volcanoes
6. Earth is unique in having plate tectonics; other bodies have stagnant lids
7. Fluvial features on Mars: strong evidence for past liquid water
8. Cryovolcanism on Enceladus, Europa, Triton: liquid water beyond the habitable zone

Looking ahead to Lecture 8

- This lecture read planetary surfaces as a record: craters date them, volcanism and tectonics build them, erosion wears them down.
- Lecture 8 goes underneath, to the interiors that drive the endogenic half of that record: seismology, moment of inertia, and the layered structure they reveal.
- The stagnant lid and the plate-tectonic regime return there as consequences of interior heat transport rather than as surface styles.

Questions?

Next: Lecture 8, Planetary interiors: structure, composition & dynamics

Thu 1 Oct, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 2 Oct, 13:00–15:00, room 5419.0161 (Collegezaal)

`https://ips.formingworlds.space`